

Leonardo Herrera

Cell Phone: (281) 900-6047

Spring, TX

Portfolio: herrera2.github.io/Personal-Website/index.html

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LinkedIn: linkedin.com/in/lh-mech-eng

Skills

SolidWorks, Creo Parametric, Siemens NX, AutoCAD, GD&T, Tolerance Stack-Up, Sheet Metal Design, CNC Nesting, MATLAB/Simulink, System Identification, Transfer Function Analysis, FEA, Hardware-in-the-Loop Simulation, Systems Integration, Sensor Calibration, Root Cause Analysis, Process Improvement, Trilingual (English, Spanish, Portuguese)

Work Experience

Manufacturing Design Engineer

May 2025 – Present

Duke Manufacturing – CounterCraft

Houston, TX

- Designed and released ~70 custom sheet-metal components daily in SolidWorks for made-to-order commercial countertop systems, contributing to the delivery of 100+ custom counter assemblies over a 10-month period.
- Generated flat patterns, bend deductions, and CNC-ready nesting programs for ~200 parts per day, improving material utilization, reducing scrap, and maintaining high-throughput manufacturing schedules.
- Developed and implemented a parametric universal mullion framework in SolidWorks that standardized internal support structures across product lines, reducing modeling and design iteration time by ~15–20%.
- Collaborated with fabrication, CNC, and production teams to resolve fit-up and structural nonconformances, implement corrective design changes, and improve manufacturability and shop-floor efficiency.
- Interpret engineering drawings in AutoCAD to ensure compliance with manufacturing and quality standards.

Mechanical & Test Engineering Intern

March 2022 – September 2023

NASA Johnson Space Center – CACI (Contractor), PVAMU (Subcontractor)

Houston, TX

- Supported spacecraft docking certification campaigns comprising 300+ dynamic test cases on the [SDTS](#), contributing to hardware setup, calibration, and execution of real-time hardware-in-the-loop simulations.
- Integrated and configured electromechanical instrumentation, optical tracking systems, and laser metrology equipment for complex multi-system test operations supporting full test-cycle execution and validation activities.
- Operated a 6DOF robotic system in closed-loop simulations, where real-time sensor feedback informed physics-based models to dynamically adjust motion during docking events. Interpreted sensor and system data during test execution.
- Designed and improved test hardware, including sensor mounts and structural interfaces, using Creo Parametric and Creo Simulate to improve measurement reliability, hardware integration, and test repeatability.
- Developed calibration tooling and validated force measurement systems, improving data accuracy and repeatability.
- Built dynamic system models in MATLAB/Simulink, applying system identification and transfer-function analysis to characterize robotic system behavior and support control-filter assessment studies.
- Collaborated with multidisciplinary engineering teams supporting mission-critical validation efforts for international and commercial partner vehicles, including SpaceX and Japan Aerospace Exploration Agency.
- Recognized with two NASA Elite Team Awards and two CACI Recognitions for contributions to spacecraft docking simulation and SDTS test operations.

Engineering Projects

Robotic Arm Simulation (NASA Trick) (*In Development*)

May 2026 – Present

- Developing 3-DOF robotic arm simulation in NASA Trick (Python/C++); implementing forward kinematics and inverse kinematics framework for real-time control and visualization

Stewart Platform Kinematics & Simulation (stewart_platform.html)

March 2026 – April 2026

- Developed inverse/forward kinematics for a 6-DOF Stewart platform; solved constrained actuator lengths via numerical optimization and enabled real-time 3D motion visualization

Education

Master of Science in Engineering

Graduated December 2023

Prairie View A&M University, Prairie View, TX

GPA: 3.80

- Thesis: Control Filter Assessment on Six Degree of Freedom Stewart Platform (Publication Pending)

Bachelor of Science in Mechanical Engineering

Graduated May 2022

Prairie View A&M University, Prairie View, TX

GPA: 3.67

- Senior Project: Designed, modeled, and built a prototype of a Lightweight Deployable Shield for Ball Aerospace.